

LEARNING PLAN

COMPUTER NETWORK SETUP MAY-JULY 2026

Institution Name	ABERDARE TECHNICAL COLLEGE
Department	ICT Department
Course Title	Information Communication Technology
Unit of Competency	Computer Network Setup
Name of the Trainer	Mr ndirangu
Class/Level	5
Academic Year	2026

Skill or Job task:

Terminate network cables, connect network cables and perform computer network Maintenance.

Bechmark Criteria to be used:

1. Terminate Computer network cables

1.1 Network devices are selected as per technical requirements

1.2 Network Cable trunking is performed as per work requirements

1.3 Network Cable termination is performed as per work requirements

2. Connect Computer network cables

2.1 Safety measures are observed as per workplace procedure.

2.2 Setting up Network devices is performed as per work requirements

2.3 Network Cable testing is performed as per work requirements

2.4 Network Cable connection performed as per networking standards

2.5 Network connection is established as per networking standards

2.6 Network testing is performed as per work requirements

3. Perform Computer Network Maintenance

3.1 Computer network is monitored as per work requirement

3.2 Computer network troubleshooting is performed as per networking standards

3.3 Computer network is optimized as per networking standards.

CONTENTS

Week	Session/Topic	Learning Outcomes By the end of this session, the trainee should be able to:	Trainer Activities	Trainee Activities	Resources/Tools	Assessment Methods
Week 1	Terminate Computer Network Cables	• Define computer networking concepts correctly. • Describe at least three benefits of computer networking with 90% accuracy. <i>(Remembering & Understanding)</i>	Explain networking concepts using examples and demonstrations. Guide class discussions on benefits of networking.	Listen, take notes, participate in discussions, and answer questions.	Computers, projector, charts	Oral questions, quiz
		• Compare different network topologies correctly. • Draw and identify common network topologies accurately. <i>(Understanding & Analyzing)</i>	Demonstrate network topology diagrams and explain their applications.	Draw and identify network topologies in groups.	Whiteboard, topology charts	Assignment, presentation

Week	Session/Topic	Learning Outcomes By <i>the end of this session, the trainee should be able to:</i>	Trainer Activities	Trainee Activities	Resources/Tools	Assessment Methods
Week 2	Terminate Computer Network Cables	• Identify networking devices correctly. • Classify computer network components according to their functions. (<i>Remembering & Applying</i>)	Display networking devices and explain their functions.	Observe, identify, and classify networking devices.	Routers, switches, NICs, cables	Practical observation
		• Select appropriate networking tools for cable installation tasks correctly. (<i>Applying</i>)	Demonstrate networking tools and explain their uses.	Practice selecting and handling networking tools.	Crimping tool, stripper, tester	Practical exercise
Week 3	Terminate Computer Network Cables	• Arrange network cable wires according to T568A and T568B standards correctly. (<i>Applying</i>)	Demonstrate cable colour coding standards.	Arrange cable wires following demonstrated standards.	UTP cables, RJ45 connectors	Practical assessment
		• Explain cable trunking correctly. • Differentiate various cable trunking	Explain cable trunking methods and safety considerations.	Observe demonstrations	Trunking samples, charts	Written exercise

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		methods accurately. <i>(Understanding & Analyzing)</i>		and identify trunking methods.		
Week 4	Terminate Computer Network Cables	• Use cable trunking tools safely and correctly during installation. <i>(Applying)</i>	Demonstrate proper use of trunking tools and safety procedures.	Practice using trunking tools under supervision.	Measuring tape, drill, screwdriver, spirit level	Observation checklist
		• Describe the cable termination process correctly. <i>(Understanding)</i>	Explain cable termination procedures step by step.	Take notes and participate in guided discussions.	Network cables, projector	Quiz
Week 5	Terminate Computer Network Cables	• Demonstrate proper handling of cable termination tools safely. <i>(Applying)</i>	Demonstrate safe handling of termination tools.	Handle tools and practice safety procedures.	Crimping tool, cable cutter, wire stripper	Observation
		• Terminate a straight-through cable successfully within 15 minutes	Demonstrate cable stripping, colour	Perform cable termination practical activities.	UTP cables, RJ45 connectors	Practical test

Week	Session/Topic	Learning Outcomes By the end of this session, the trainee should be able to:	Trainer Activities	Trainee Activities	Resources/Tools	Assessment Methods
		following networking standards. (<i>Creating</i>)	coding, and crimping procedures.			
Week 6	Connect Computer Network Cables	• Apply workplace safety procedures during networking activities consistently. (<i>Applying</i>)	Explain networking safety rules and demonstrate use of PPE.	Wear protective equipment and follow safety procedures.	Gloves, dust coats, boots, first aid kit	Observation
		• Configure network devices correctly. • Connect network devices in a small LAN setup successfully. (<i>Applying & Creating</i>)	Demonstrate setup of routers, switches, and access points.	Connect and configure networking devices practically.	Routers, switches, access points	Practical assessment
Week 7	Connect Computer Network Cables	• Explain cable testing methods used in networking correctly. (<i>Understanding</i>)	Explain cable testing methods using demonstrations.	Observe demonstrations and explain testing procedures.	Cable testers, charts	Quiz
		• Perform cable testing procedures correctly. •	Demonstrate continuity testing and	Conduct cable tests and identify	Multimeter, cable tester	Practical exercise

Week	Session/Topic	Learning Outcomes By <i>the end of this session, the trainee should be able to:</i>	Trainer Activities	Trainee Activities	Resources/Tools	Assessment Methods
		Detect cable faults accurately using testing devices. <i>(Applying & Analyzing)</i>	fault detection procedures.	cable faults practically.		
Week 8	Connect Computer Network Cables	• Operate cable testing tools effectively and safely. <i>(Applying)</i>	Demonstrate operation of testing tools.	Practice using testing tools independently.	Cable tester, wire stripper, cutter	Observation
		• Explain networking standards correctly. • Relate networking standards to communication systems accurately. <i>(Understanding & Analyzing)</i>	Explain networking standards using practical examples.	Participate in discussions and answer questions.	Projector, notes	Assignment

Week	Session/Topic	Learning Outcomes By <i>the end of this session, the trainee should be able to:</i>	Trainer Activities	Trainee Activities	Resources/Tools	Assessment Methods
Week 9	Connect Computer Network Cables	• Establish network connections successfully in a simulated environment. (<i>Applying & Creating</i>)	Guide trainees in establishing network connections.	Setup and test network connections in groups.	Computers, switches, cables	Practical project
	Perform Computer Network Maintenance	• Describe the importance of network monitoring and maintenance correctly. (<i>Understanding</i>)	Explain maintenance procedures and monitoring practices.	Take notes and participate in discussions.	Projector, notes	Written exercise
Week 10	Perform Computer Network Maintenance	• Inspect network hardware using maintenance tools correctly. • Monitor network devices effectively. (<i>Applying & Evaluating</i>)	Demonstrate network monitoring procedures.	Monitor and inspect networking devices practically.	Cable testers, crimping tools	Observation

Week	Session/Topic	Learning Outcomes By <i>the end of this session, the trainee should be able to:</i>	Trainer Activities	Trainee Activities	Resources/Tools	Assessment Methods
		• Analyze network connectivity problems accurately. • Troubleshoot computer network faults successfully. (<i>Analyzing & Evaluating</i>)	Provide troubleshooting scenarios and guide learners.	Diagnose and solve network faults practically.	Routers, switches, cables	Troubleshooting task
		• Recommend suitable network optimization techniques correctly. • Implement network optimization solutions for improved performance. (<i>Evaluating & Creating</i>)	Demonstrate network upgrading and optimization techniques.	Upgrade cables/devices and optimize network setup.	Network devices, upgraded cables	Final practical project